

DS Assignment 6B

Implement Bully and Ring algorithm for leader election.

//Ring.java

```
import java.util.*;

public class Ring {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of processes: ");
        int n = sc.nextInt();

        int[] id = new int[n];
        boolean[] alive = new boolean[n];

        for (int i = 0; i < n; i++) {
            System.out.print("Enter ID for process " + i + ": ");
            id[i] = sc.nextInt();
            alive[i] = true;
        }

        Arrays.sort(id);
        int coordinator = id[n - 1];

        System.out.println("Initial Coordinator: Process " + coordinator);

        while (true) {
            System.out.println("\n1.Start Election 2.Crash 3.Exit");
            int ch = sc.nextInt();
        }
    }
}
```

```

if (ch == 1) {
    System.out.print("Enter initiator index: ");
    int init = sc.nextInt();

    if (init < 0 || init >= n || !alive[init]) {
        System.out.println("Invalid initiator.");
        continue;
    }

    int max = id[init];
    int current = (init + 1) % n;

    System.out.println("Election started by Process " + id[init]);

    while (current != init) {
        if (alive[current]) {
            System.out.println("Message passed to Process " + id[current]);

            if (id[current] > max)
                max = id[current];
        } else {
            System.out.println("Process " + id[current] + " is dead, skipped.");
        }

        current = (current + 1) % n;
    }

    coordinator = max;
    System.out.println("New Coordinator: Process " + coordinator);
}

```

```
} else if (ch == 2) {  
    System.out.print("Enter index to crash: ");  
    int p = sc.nextInt();  
  
    if (p < 0 || p >= n || !alive[p]) {  
        System.out.println("Invalid process.");  
        continue;  
    }  
  
    alive[p] = false;  
    System.out.println("Process " + id[p] + " crashed.");  
  
    if (id[p] == coordinator)  
        System.out.println("Coordinator crashed. Start election.");  
  
} else if (ch == 3) {  
    break;  
  
} else {  
    System.out.println("Invalid choice.");  
}  
}  
  
sc.close();  
}  
}
```

Output –

javac Ring.java

java Ring

Enter number of processes: 4

Enter ID for process 0: 4

Enter ID for process 1: 7

Enter ID for process 2: 2

Enter ID for process 3: 9

Initial Coordinator: Process 9

1.Start Election 2.Crash 3.Exit

2

Enter index to crash: 3

Process 9 crashed.

Coordinator crashed. Start election.

1.Start Election 2.Crash 3.Exit

1

Enter initiator index: 0

Election started by Process 2

Message passed to Process 4

Message passed to Process 7

Process 9 is dead, skipped.

New Coordinator: Process 7

1.Start Election 2.Crash 3.Exit

3